

Answer Key & Teacher Notes

Session 1 – Distance Between Points on a Grid • Session 2 – Comparing Distances on the Plane

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Distance Between Points on a Grid

TALK PROMPTS

- How can the coordinates alone tell you whether a trip runs across or up and down?
- Why does a trip that crosses the x-axis join the two distances instead of taking one away?

PARTNER TASK

Form A, problem 1. One partner finds the distance from (3, 14) to (3, 4) (part a); the other finds the distance from (–11, 2) to (6, 2) (part b). Each explains whether they added or subtracted, then they switch and check.

WATCH FOR

- Subtracting the absolute values when the points are on opposite sides of an axis ((2, –9) to (2, 6) as 3 instead of 15).
- Using the coordinate that stays the same instead of the one that changes.
- Changing the y-coordinate for a move left or right when finding a missing point.

SESSION 2 • TEACHER NOTES

Comparing Distances on the Plane

TALK PROMPTS

- How do the signs of the changing coordinates tell you whether to add or to subtract?
- How can the coordinates alone tell you which ride crosses an axis?
- Why must both distances use the same scale before you compare them?

PARTNER TASK

Form A, problem 1. Both trips start at (5, –3). One partner finds the distance to (5, –15) (part a); the other finds the distance to (5, 10) (part b). Compare: who added, who subtracted, and why? Then decide which trip is longer.

WATCH FOR

- Subtracting absolute values for points on opposite sides of an axis ((–8, 3) to (4, 3) as 4 instead of 12).
- Comparing grid units for one trip with scaled meters for the other trip.
- Writing a negative distance, such as –9, instead of a length.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.C.8	1	2	1	1	2	2	2	3	3	1	1
S1 • Form B	6.NS.C.8	1	2	1	1	2	2	2	3	3	1	1
S2 • Form A	6.NS.C.8	1	2	2	2	2	2	2	3	3	1	1
S2 • Form B	6.NS.C.8	1	2	2	2	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

distance: How far apart two points are. It is never **negative**.

absolute value: How far a number is from **0**, ignoring its sign.

horizontal distance: Distance side to side. Use the **x**-coordinates.

vertical distance: Distance up and down. Use the **y**-coordinates.

- If two points share a y-coordinate, find the distance with the **x**-coordinates.
- If the changing coordinates have opposite signs, **add** their absolute values. If they have the same sign, **subtract** them.

Example: The points share the y-coordinate **3**, so use the x-coordinates. -4 and 5 have **opposite** signs, so the trip crosses the y-axis. Add the absolute values: $4 + 5 = 9$.

Answer: The distance is **9** units.

YOU TRY

1. a. $6 + 3 = 9$ units b. $7 - 2 = 5$ units
2. B = (3, 4)

EXIT TICKET

1. a. $3 + 6 = 9$ units b. $8 - 2 = 6$ units
2. B = (-2, -2)
3. B (horizontal line)

PRACTICE FORM A

1. a. $14 - 4 = 10$ b. $11 + 6 = 17$ c. $9 + 6 = 15$ d. $6 + 5 = 11$
2. a. L(-8, 5), P(7, 5), C(-8, -6) b. same y; $8 + 7 = 15$ km
c. same x; $5 + 6 = 11$ km
3. $13 + 4 = 17$ m
4. $20 - 7 = 13$ units
5. H = (-5, -3)
6. $18 - 7 = 11$ ft. Both y-coordinates are positive (same side of the x-axis), so subtract.
7. the x-coordinates; $17 + 12 = 29$ blocks
8. a. $4 + 5 = 9$ blocks b. $6 + 3 = 9$ blocks c. $9 + 9 = 18$ blocks
9. The y-coordinates -9 and 6 have opposite signs, so the trip crosses the x-axis. Malik should add: $9 + 6 = 15$ units.
10. B (7 units)
11. B (9 units)

Self-check word LENGTH

PRACTICE FORM B

1. a. $12 - 5 = 7$ b. $9 + 7 = 16$ c. $8 + 5 = 13$ d. $5 + 9 = 14$
2. a. M(-6, 4), U(8, 4), S(-6, -7) b. same y; $6 + 8 = 14$ km
c. same x; $4 + 7 = 11$ km
3. $11 + 5 = 16$ m
4. $20 - 8 = 12$ units
5. H = (-5, -2)
6. $17 - 8 = 9$ ft. Both y-coordinates are positive (same side of the x-axis), so subtract.
7. the x-coordinates; $15 + 13 = 28$ blocks
8. a. $5 + 6 = 11$ blocks b. $4 + 2 = 6$ blocks c. $11 + 6 = 17$ blocks
9. The y-coordinates -8 and 5 have opposite signs, so the trip crosses the x-axis. Amina should add: $8 + 5 = 13$ units.
10. B (9 units)
11. B (vertical line)

Self-check word ORIGIN

GUIDED NOTES

quadrant: One of the four **sections** of the coordinate plane.

compare: Decide which amount is **greater** or less.

scale: How much real length one grid **unit** stands for.

- Find **each** distance first. Then compare the two numbers.
- With a scale, change **both** distances to real units before you compare.

Example: A to B: same x, so $9 - 2 = 7$ units = 35 m. A to C: same y, opposite signs, so $1 + 8 = 9$ units = 45 m.

Answer: **C** is farther, by 10 m.

YOU TRY

- a. add: $2 + 7 = 9$ units b. subtract: $12 - 5 = 7$ units
- Ana: $3 + 6 = 9$ units; Ben: $4 + 6 = 10$ units. Ben walks farther, by 1 unit.

EXIT TICKET

- a. $5 + 4 = 9$ units b. $9 - 1 = 8$ units
- Lee: $6 + 3 = 9$ units. Mia: $4 + 7 = 11$ units. Mia, by 2 units.
- A** (5 units)

PRACTICE FORM A

- a. $15 - 3 = 12$ b. $3 + 10 = 13$ c. $4 + 6 = 10$ d. $8 - 2 = 6$
- Gym(-6, 4), Field(7, 4), Office(-6, -5). a. $6 + 7 = 13$ units = 130 m b. $4 + 5 = 9$ units = 90 m c. Gym to Field, by 40 m
- Tariq: $5 + 8 = 13$ blocks. Jada: $9 + 6 = 15$ blocks. Jada, by 2 blocks.
- $U = (-4, -6)$
- $M = (-7, -5)$
- Kendra: $12 - 3 = 9$ km. Owen: $2 + 10 = 12$ km. Owen, by 3 km.
- Ana: $5.75 + 3.25 = 9$ units. Ben: $9.75 - 5.75 = 4$ units. Ana, by 5 units.
- a. Stop 1: 12, Stop 2: 9, Stop 3: 4 units b. Stop 3, Stop 2, Stop 1 c. $12 + 9 = 21$ units
- The x-coordinates -8 and 4 have opposite signs, so the points are on opposite sides of the y-axis. Ines should add: $8 + 4 = 12$ units.
- B** (subtraction)
- C** (negative)

Self-check word LONGER

PRACTICE FORM B

- a. $13 - 2 = 11$ b. $2 + 8 = 10$ c. $3 + 6 = 9$ d. $9 - 1 = 8$
- Pond(-7, 5), Meadow(5, 5), Cabin(-7, -2). a. $7 + 5 = 12$ units = 120 m b. $5 + 2 = 7$ units = 70 m c. Pond to Meadow, by 50 m
- Rosa: $4 + 7 = 11$ blocks. Kofi: $8 + 6 = 14$ blocks. Kofi, by 3 blocks.
- $U = (-3, -6)$
- $M = (-6, -4)$
- Mei: $13 - 4 = 9$ km. Andre: $3 + 11 = 14$ km. Andre, by 5 km.
- Ana: $4.75 + 2.25 = 7$ units. Ben: $9.75 - 4.75 = 5$ units. Ana, by 2 units.
- a. Stop 1: 11, Stop 2: 8, Stop 3: 5 units b. Stop 3, Stop 2, Stop 1 c. $11 + 8 = 19$ units
- The x-coordinates -6 and 9 have opposite signs, so the points are on opposite sides of the y-axis. Diego should add: $6 + 9 = 15$ units.
- B** (addition)
- C** (14 units)

Self-check word METERS

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · DISTANCE BETWEEN POINTS ON A GRID

1. $A(-4, 3)$, $B(3, 3)$, $C(3, -2)$, $D(-4, -4)$. a. $4 + 3 = 7$ units
b. $3 + 2 = 5$ units c. $3 + 4 = 7$ units
2. a. 8 b. 5 c. 9 d. 11 e. 5 f. 6.5 g. 10 h. 4.5
3. Same x; $3 + 5 = 8$ blocks
4. $Q = (-4, -3)$
5. Both y-coordinates are negative (same side of the x-axis), so subtract: $6 - 2 = 4$ units.

SHEET 2 · COMPARING DISTANCES ON THE PLANE

1. $H(-4, 2)$, $L(3, 2)$, $P(-4, -3)$. a. $4 + 3 = 7$ blocks b. $2 + 3 = 5$ blocks c. the Park, by 2 blocks
2. a. $8 - 3 = 5$ b. $6 + 7 = 13$ c. $7 - 1 = 6$ d. $2.5 + 4.5 = 7$
3. Ana: $5 + 6 = 11$ units. Jay: $4 + 6 = 10$ units. Ana runs 1 unit farther.
4. Add. The y-coordinates 7 and -2 have different signs, so the path crosses the x-axis: $7 + 2 = 9$ units.
5. **A** (JK, by 2 units)
6. **C** (8 units)